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Project management for academic research projects: balancing structure and flexibility

Hélène Riol* and Denis Thuillier

Department of Management and Technology,
Université du Québec à Montréal,
P.O. Box 8888, succursale Centre-ville,
Montreal, Quebec H3C 3P8, Canada
E-mail: riol.helene@courrier.uqam.ca
E-mail: thuillier.denis@uqam.ca
*Corresponding author

Abstract: Academic research faces new methods of knowledge production that trigger a need for managing research by projects. However, the literature reports friction between management and research. In this study, we investigate whether and to what extent academic research projects can be managed using classical project management (PM) principles. An analysis of managerial facts from interviews with ten university researchers indicates that research projects are PM-compatible considering certain structural similarities and a cultural acceptance of PM value. However, the human factors and uncertainties inherent in research are not addressed by classical PM. A grounded analysis allows for modelling a PM perspective that integrates *soft* and *hard* contingent aspects equally by combining structured and flexible approaches adapted to managing projects of an exploratory, uncertain and complex nature. We thus developed a prescriptive framework for facilitating PM implementation in academic research at the institutional, organisational and operational levels.

Keywords: academic research projects; scientific research; exploratory projects; project management; research management; contingency; complexity; uncertainty; balancing project management.

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Biographical notes: Hélène Riol is a PhD in Cellular and Molecular Biology from Laval University, Quebec, Canada. Her previous research interest focuses on the molecular biology of aging. However, after a working experience as a scientific project manager in a governmental laboratory of biotechnology, she got interested in how researchers manage their projects and whether conventional project management is suitable for this particular type of project. She completed meantime a Master degree in Project Management at the University of Quebec in Montreal (UQÀM).

Denis Thuillier is an Engineer (ENSCI, Paris) and PhD in Applied Economics (Aix-Marseille III), is a Full Professor, Department of Management and Technology, at ESG-UQÀM, Montreal. He was the Director of master programmes in project management at this university. A member of the Project Management Chair (ESG-UQÀM), its recent research focus on investigating project management practices in the International Development Industry (World Bank, UNDP, Multilateral Development Banks, etc.) and on the

1 Introduction

University research traditionally aims to produce new knowledge independent of ideological, political or economic considerations. However, in the knowledge-based economy of the past 20 years, the quest in academic research to make social contributions and new research funding methods have generated a strong demand for management procedures that hold research institutes accountable for meeting their obligations, maintaining their reputation and remaining competitive in terms of their productivity (Gibbons et al., 1994; Kirkland, 2008).

The new direction in research management centres primarily on a management-by-project strategy based on project management (PM) principles (Kirkland, 2008). Typical approaches (thereafter classical PM) provide a framework centred on project life cycle and rely on specific skills, processes and tools (PMI, 2008) to transform project resources into an efficient and productive system (Dinsmore and Cooke-Davies, 2006).

However, in the universities where management-by-project was implemented, conflicts quickly surfaced. Perry (2006) suggests that primary cause owed to the fact that implementation processes were *insensitive* to the academic context and failed to take account of scientist-specific skills and culture. They may also have been inappropriate for managing research projects. Indeed, the *inadequate format and intensity* of PM processes at the intangible *soft* cultural and organisational levels as well as the more tangible and technical *hard* levels are counterproductive and could hinder the success of projects (Shi, 2011). Few empirical investigations have examined the relevance of classical PM in managing *academic* research projects, particularly with respect to the scientists involved. The objective of this study is *to determine whether and to what extent academic research projects can be adequately managed using classical PM principles, processes and tools*. We will therefore evaluate the compatibility of PM processes and tools with *soft* and *hard* research systems using an engineering approach to research-action.

Next, we present our conceptual framework, research questions and research methodology, followed by our results and a discussion of their implications (theoretical and managerial). We conclude with an examination of limits and future research opportunities.

2 Conceptual framework and research questions

Scientists produce new knowledge using various approaches according to the context. However, what follows concerns only academic research in the sciences (fundamental or applied, natural, human or social) without regard for epistemological and methodological considerations.

The scientific research cycle includes five phases (idea conception, research plan, plan execution, dissemination of findings and project closure), which resemble the phases of a project. Furthermore, scientific research comprises a temporary endeavour that

assembles resources in order to deliver a unique output (new knowledge) subject to specific ‘quality’ constraints and within a budget. This is termed a ‘project’ (PMI, 2008). If research projects are projects, which seems a reasonable assumption, it also seems logical to ask whether (and to what extent) classical PM processes could apply to them.

2.1 *The socio-economical context of academic research*

Academic research is supported by government subsidies, which allows some independence with regards to ideological, political or economic considerations. Research is a built-in task for university professors. They may choose their research programme with a fair degree of autonomy, but peers will evaluate the creativity and originality of their contribution. This is the traditional way of producing knowledge (‘mode 1’), a sort of ‘science for scientists’ (Gibbons et al., 1994). Although this approach still predominates, the situation has evolved in the last two decades.

Institutional subsidies and grants have steadily decreased and growing competition is emerging. Like project managers, the principal investigators (PI) involved in research projects must now account for the strategic orientations of institutional funding organisations and private sponsors. This forms part of the current trend toward ‘utilitarianism’ in a knowledge-based economy, where science must support national economic and social development, contribute to improving competitive advantage and generate high returns for society (Perry, 2006). This ‘mode 2’ knowledge production (Gibbons et al., 1994), which combines ‘*excellence with relevance*’ (Perry, 2006), has triggered the rise of research management strategies causing some discomfort and counter productivity (Leese and Storey, 2005; Van Elden, 2010). With respect to PM, especially the classical approach to it, one may reasonably wonder whether such processes, tools and techniques are appropriate in managing research projects. Are we facing an incompatibility between PM practices and research culture and values (Barré, De Laat and Theys, 2007) or simply inadequacies in PM tools and techniques for monitoring research activities? Furthermore, how influential is contingency in explaining the tensions between academic research and management (Perry, 2006)? Although these questions have been addressed (Ernø-Kjølhed et al., 2001; Sousa and Hendriks, 2008), to the best of our knowledge, there appears to be no *empirical* documentation on the management practices of researchers or whether such practices are ‘PM-compatible’.

2.2 *PM: soft and hard aspect considerations*

PM has its own corpus of knowledge, processes and tools. The PMBOK (PMI, 2008) identifies five groups of processes that support project initiation or identification, planning, implementation, control and closure in each of nine knowledge areas (integration, scope, time, cost, quality, human resources and stakeholders, communications, risks and procurement) through project life cycle management.

PM is valuable: it increases client and stakeholder satisfaction, lowers costs and improves productivity (Thomas and Mullaly, 2007). A corpus of knowledge such as the PMBOK conveys a universal, standardised and normative vision of PM. Nevertheless, a project’s success depends on more than simply following ‘state of the art’ PM practices. The implementation of PM processes in a particular organisation or for a specific type of project is subject to contingencies (Dinsmore and Cooke-Davies, 2006). Several ‘atypical’ PM approaches have grasped this reality, such as those relying on project

typologies (Sausser et al., 2009; Boehm and Turner, 2003), those aimed at managing project complexity (Cooke-Davies et al., 2007) and agile methods for information technology projects (Fernandez and Fernandez, 2008, 2009).

Nonetheless, we draw attention below to two factors, ‘soft’ and ‘hard’ aspects, that should be considered for successful PM implementation (Shi, 2011). *Soft* aspects refer to the capacity and natural willingness of an organisation to integrate PM principles into its practices. *Hard* aspects refer to the adequacy of PM tools and techniques in light of the managerial processes and tools already used in an organisation. In the academic context, *soft* aspects are reputedly challenging to manage (Ernø-Kjølhed et al., 2001). First of all, it seems appropriate to ask how much value researchers place on managerial processes versus scientific methods, since their priority is to deliver valuable new knowledge. Secondly, research teams differ from traditional project teams in ways that cannot be ignored. For example, under the supervision of a tenured professor, the standard research team primarily consists of contractual junior scientist trainees (students) who require coaching (Lafrance, 2009). Furthermore, research projects are implemented by a professor-student duo (each student being in charge of a ‘sub-project’) instead of a team.

With respect to *hard* aspects, one may wonder whether the managerial activities of scientists spontaneously converge with or diverge from classical PM principles for legitimate reasons linked to the characteristics of research projects. Indeed, uncertainty forms part of such projects, which make them difficult to plan *ex ante*.

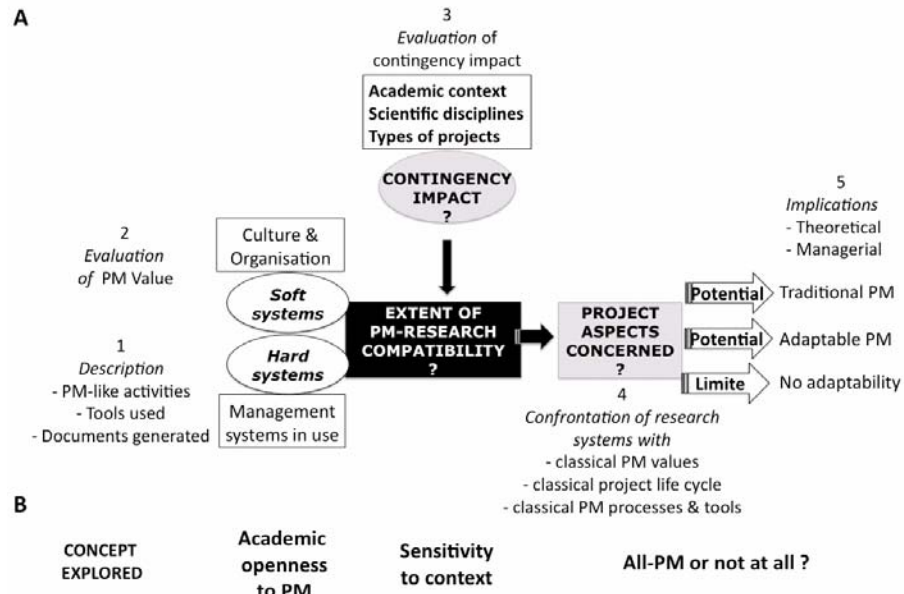
Classical PM emphasises well-structured, fully pre-planned projects that can be controlled to deliver the expected results. Newer PM approaches recognise its value, but as Boehm and Turner (2003) put it, they also support the idea of balancing traditional methods with agile, flexible approaches to managing contemporary projects. Whether classical PM can accommodate *soft* and *hard* research PM, entirely, partially or not at all, is the focus of our investigation.

2.3 Conceptual framework

The above considerations lead to our conceptual framework (Figure 1). Based on academic openness to PM, we determined the compatibility of processes and tools (*hard* side) by studying the way that PIs manage, the tools they use, and the documentation they produce. The PIs’ perception of the value of *PM-like* processes constitutes the *soft* side. We then introduced contingency factors (mainly academic context, scientific disciplines and project particularities and constraints) and their impact on PM-research compatibility (sensitivity to the context concept; Perry, 2006). The analysis should allow us to identify the conditions underlying PM practices and value, and their limitations (‘all-PM or not at all’ concept), in order to make a theoretical and managerial contribution concerning PM potentiality in this particular area of non-traditional industry.

We have therefore investigated the following questions:

- Are classical PM processes and tools compatible with the management of academic research projects?
- What are the impacts of academic-based contingent factors?
- Which aspects, phases or activities of academic research projects are PM-compatible?

Figure 1 (a) Conceptual framework and (b) Theoretical concepts investigated

3 Methods

3.1 Research-action strategy

Research-action traditionally aims to produce knowledge *through* a transformation, based on a reflexive approach involving both the researchers and field participants. The 'engineering approach' to research-action differs, however, from the traditional approach by generating theoretical knowledge *in view* of this transformation (Allard-Poesi and Perret, 2003). It aims first to understand and build the issue confronting the field by comparing theoretical knowledge with field actor practices. Secondly, this issue is translated into a grounded, explanatory and/or prescriptive model that is tested in a third step. Since it involves practitioner collaboration, this strategy (particularly the first two steps) appears to be the one best suited to the present study. According to Cicmil et al. (2006), it allows an understanding of how *soft* social processes, combined with contingent factors and *hard* project characteristics, can impact practitioner actions and perceptions of PM value. Accordingly, the first step is expected to clarify the implicit *PM-like* practices of scientists (*hard* side), taking into account their environment (contingent factors) and 'context-dependent judgement' of PM value (*soft* side; Cicmil et al., 2006). In the second step, a grounded model is deduced with concern for both the PM practitioner and academic communities (managerial and theoretical contributions).

3.2 *Participants*

Nine university professors and one experienced research professional (six men and four women; hereafter called ‘PI’), all performing similar functions (teaching, research and administrative tasks), were recruited on a voluntary basis at the Université du Québec à Montréal (UQÀM) and McGill University (Canada). Since these universities have not implemented management by project, we were able to explore their experience-based practical knowledge (Larose, 2006). This study encompasses social sciences (political science, sociology and geography; four participants) and biological sciences (six participants) that use field or experimental approaches, or a combination of both, in order to understand their respective impact on researchers’ *PM-like* practices. The participants have from 21 months to 33 years of experience in their current positions, and research team sizes vary between 4 and 15 members.

3.3 *Data collection*

Data were collected in three phases:

- 1 a preliminary study involving six researchers with whom we conducted unstructured or semi-structured interviews (two series of 60-minute interviews) to establish the study’s relevance, determine the field issue and improve our questionnaire
- 2 a second phase during which we completed the study by contacting four additional participants (two series of 60-minute, semi-structured interviews)
- 3 a group meeting (90 minutes) at the end of study after completing the data analysis with participants from the first two phases (four biologists and two social scientists including the two members of the research team).

This strategy allowed us to discuss and validate our observations with the participants and gather their PM concerns and suggestions.

The semi-structured interview topics included:

- 1 a description of their project context
- 2 a description of their *PM-like* activities
- 3 their evaluation of the impact of academic-based contingent factors
- 4 their evaluation of PM value.

The first two topics aimed to obtain information on *hard* research systems and the last two, on *soft* aspects, by gathering the researchers’ opinions. Throughout the interviews, to compare PM theoretical knowledge with the researchers’ practical experiences, we introduced them to classical PM processes and tools.

3.4 *Data analysis*

The interviews were digitally sound recorded with the participants’ authorisation and transcribed for data analysis purposes. We began by developing a *descriptive analysis* to gain insight into the extent of PM compatibility with *hard* and *soft* research systems (questions 1 and 2).

We then performed a qualitative content analysis using the *grounded theory* (Glaser and Strauss (1967), in Flick et al., 2004). With systematic and constant comparisons of the researchers' verbatim, the objective was to evoke *patterns* or *PM perspectives* of value to research PM, but also of potential applications in other areas or industries. This allowed us to answer questions 2 and 3.

4 Results

4.1 Descriptive analysis

4.1.1 PM processes and tools in an academic context

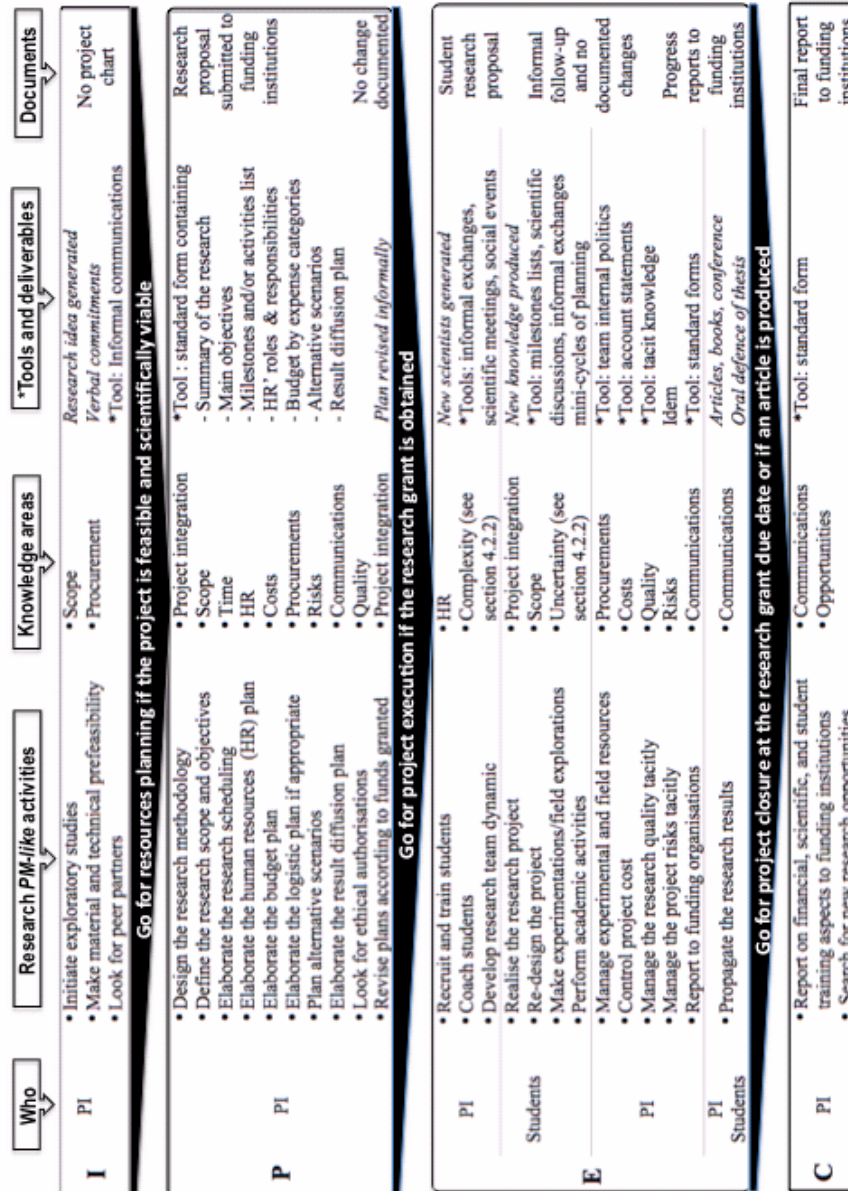
We structured PIs' *PM-like* activities according to the four phases of a classical project life cycle (Figure 2), assuming that the *structural organisation of research projects is compatible with conventional projects*. We also described the researchers' activities in terms of PM processes according to the nine PMBOK knowledge areas, suggesting that *the management activities involved in research projects include at least all activities involved in conventional projects*. This underscores that *researchers perform project manager tasks* in tandem with scientific tasks.

In summary, PIs carry out initiation and planning phases, respectively, the project pre-feasibility and feasibility. Then, master's or doctoral students implement the project. Report writing and paper publishing also take place during the implementation phase. The publication of a scientific paper or student graduation induces partial project closure. PIs theoretically close projects on the completion date set by research funding organisations. However, new research opportunities can reinitiate the project cycle.

At a first glance (Figure 2), *PIs do not use traditional, popular PM tools* such as scope and scheduling tools [work breakdown structure (WBS), Gantt bar chart, critical path method (CPM)], management software (MS Project), time and cost performance assessment tools (e.g., earned value method), risks assessment tools, etc. Instead, they use intuitive (related to tacit knowledge), simplified (e.g., activity and milestone lists) and informal (verbal) tools. For example, PIs manage risk and quality tacitly, based on their experience, intuition and judgement. Work is monitored in the form of debates about the significance of results, and feedback substitutes for control. In biology, emergent plans are discussed informally with students, and PIs may draw or summarise the main ideas on a piece of paper for inexperienced students. However, formal PM documents are minimal, and those kept are mainly used for communications with external stakeholders (research institutions and funding organisations) particularly in the planning (PIs' research proposal), implementation (students' research proposal) and closure (PIs' final reporting) phases. In the event of insufficient funding, which occurs almost systematically, PIs must modify the project's scope, objectives, etc., although these modifications are not updated in writing but are made informally. These findings may indicate that *classical PM-like tools are not used because researchers are not aware of them*, a plausible explanation since the PIs did not receive any PM training. Moreover, these tools might not work effectively in an academic context, as reflected by the spontaneous, *more intuitive and flexible* in-house *PM-like* tools used by scientists. Therefore, although we observed some compatibility between traditional PM and *hard* research systems (life cycle, general framework and areas of management), *contextual*

elements may account for differences in the way that researchers conduct their PM-like processes. To better understand these aspects, the two following sections report respectively, the PIs' description of their activities (specifically during the execution or implementation phase) and the PIs' perception of PM value.

Figure 2 Research project life cycle



Notes: I: initiation; P: planning; E: execution; C: closure

4.1.2 Processes in the implementation phase

PIs consider this the true 'PM phase' since the students who mainly compose the research teams 'enter the game':

"We make pre-management when we are alone in front of our computer to assemble these grant applications... because you do not have your people yet".

This phase follows the five classes of processes common to conventional projects. It begins again with initiation and planning processes to incorporate students, considered junior scientific partners rather than employees:

"It is necessary that you can adapt your projects, to assign them according to their expertise to create diversity, but also so that it corresponds to their interests...."

This calls attention to *each student's specific interaction with the PIs' overall project*, since they also have to write an individual research proposal which is evaluated and monitored by a university supervisory committee. The following is a description of a few contextual characteristics of the implementation processes.

4.1.2.1 Time and scope management

From PIs' point of view, time is the main constraint in research:

"There is the time variable for our CV, it is necessary to have a paper published each year. There is the time variable for grant renewal. Then, there is the time variable for the students..."

PIs report that they always having a time schedule *in the back of their minds* because they must cope on a daily basis with the university's 'multitasking' requirements (1/3 research, 1/3 teaching and 1/3 administration). However, time is managed more or less informally, depending on the project type, scope and the extent of initial planning. In experimental sciences, where projects are broken by objectives, PIs think that students should bear in mind the following schedule:

- 1 the completion date of their master's or doctoral thesis
- 2 the time they engage in their coursework
- 3 the date they start to write.

Everything else depends on the results, which may require changes in the initial objectives and hypothesis. Since plans emerge while such projects are in progress, *both time and scope adjustments are performed informally*. We note that, whether in social sciences or biology, field investigations are planned with more rigor than experimental investigations, which involve explorations that require more flexibility.

4.1.2.2 Human resources management

All of the PIs considered human resources management as critical. Student training is one of the two objectives of research, the other being to produce new scientific knowledge.

PIs closely coach their students, using frequent informal (verbal) exchanges and one-on-one feedback. Formal scientific meetings are important in theory but, with few exceptions, they occurred in a very flexible manner or were abandoned over time. We

therefore notice a general team development issue, given that researchers are not well-equipped to foster team dynamics and capitalise on collective learning capabilities. In this respect, none of the research teams has collective objectives or processes for evaluating team performance, as recommended in classical PM:

“Basically, what the students come for, it is a master’s or doctoral degree. Thus their first concern is their project and the finality of their training. We can set up collective objectives, but I do not know how a personal objective and a common objective would co-habit...”

4.1.3 PM value for PIs

In general, PIs recognise the relevance of PM in implementing their research projects, since poor research management can affect a career. For example, poor cost management and exaggerated perfectionism lower scientific performance and publication, whereas excessive control keeps students away. For one PI, scientific methodologies are not ‘global’ enough to carry out projects. There is also a need to look ahead and ‘see the calendar coming’. Another PI had never considered PM before, but realised that, in his present situation (many students and responsibilities apart from research), a more structured management approach could make it possible to accomplish more in less time. However, another stressed the need to maintain *a certain level of flexibility*. All mentioned the multiple talents that PIs, like traditional project managers, must demonstrate as a stand-in accountant, psychologist, human resource administrator, writer, etc. Therefore, most PIs showed a surprisingly positive attitude toward PM *for themselves*. Nonetheless, they expressed concerns *especially in the social sciences*, towards management practices *at the expense of science*:

“All that is regarded as techniques of management or management itself is considered being a kind of coercive framing. A kind of regulating technique. I am echoing the criticism that one would find in social sciences. Management science: all what it means is to optimise the profits... to force people into a mould... one knows this sort of music!”

Confronted with traditional PM tools and techniques, several researchers indicated that a simplified version of the Gantt chart had tremendous potential to help students develop their self-organisational skills and to monitor work more effectively, particularly for large research teams. It should be noted that PIs with collective infrastructures such as laboratories wanted to improve PM within their research team, and strongly endorsed the need for such PM tools. Indeed, they reported that providing well-balanced supervision (between autonomy and control) to each student is difficult and depends on several factors such as team size, tenure criteria and responsibilities other than research.

“I would like to find a way to letting them think by themselves, but at the same time, without putting my research at risk!”

As already mentioned, techniques related to collective learning and team development, although valuable, seem difficult to implement:

Finally, PIs in both the social sciences and biology find that PM training (but which one?) would be very useful to newly appointed junior researchers:

“When you are newly employed, the only meeting that you have, it is to explain that you must teach, do research and accept management assignments within your university... But, you never have a specific training on how to manage the scientific staff... I find that it is missing indeed... You learn by doing while speaking with your colleagues...”

On the whole, the descriptive analysis suggests compatibility between PM principles and both *hard* and *soft* research systems. However, the absence of PM knowledge (particularly in team management) and contextual factors explain the level and extent of the use of *PM-like* processes by PIs.

4.2 Grounded analysis

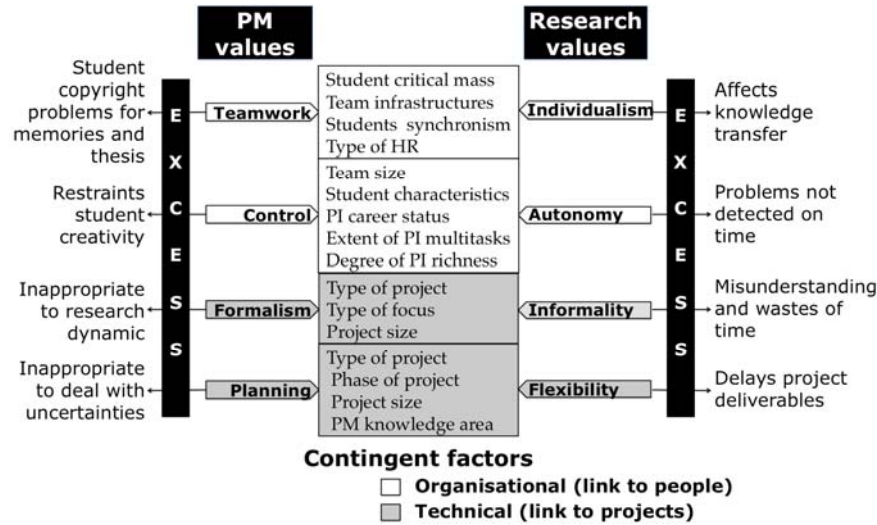
The descriptive analysis provides insight into the *PM-like* activities that researchers currently use to manage their projects and teams. The grounded analysis confirms that contingencies strongly affect their managerial style. It also reveals that *planning* is a crucial aspect of research PM. We have tentatively modelled a research PM perspective that combines these two dimensions, which provide an explanatory vision of PM for exploratory projects and, more generally, for projects that entail complexity and uncertainty.

4.2.1 The contingency dimension

Three conclusions emerge:

- 1 two types of factors, organisational (human and social) and technical (reflecting research characteristics), influence researchers' managerial strategies
- 2 depending on the context, researchers favour a more structured and/or flexible PM, given the need to arbitrate between classical (structured) *PM-like* values and those (flexible) valued in research
- 3 in the current socio-economic research context, researchers must strike a dynamic balance between structure and flexibility to manage their projects because any imbalance will lead to problems.

Figure 3 supports these conclusions. It shows that researchers must arbitrate between teamwork and individualism, student control or autonomy, formality and informality as well as planning and flexibility, in order to allow both efficient PM and creative research.

Figure 3 Impact of contingent factors on PIs' managerial styles

Notes: Depending on the factor and its particular context of realisation, PIs favour classical *PM-like* or traditional research values. For example, the type of HR (human resources): teams composed mainly of professionals of research are more prone to teamwork than those with only students since students are evaluated on an individual basis for their master or doctorate degree.

4.2.2 The planning dimension

Four conclusions emerge:

- 1 some situations (socio-economical and related to project characteristics) require planning for better *research productivity and efficiency*, while others (human and related to project characteristics) demand the flexibility needed to allow for *creativity and learning as well as exploration and changes during project implementation*
- 2 *soft* organisational (social and human) research systems require a balancing of planning and teaching (and learning) *values*, in order to train and support student creativity *while remaining productive*
- 3 *hard* technical (project characteristics) research systems require a balancing of project planning and flexibility *needs* in order to support project productivity *and* exploration
- 4 researchers must balance structure and flexibility to satisfy both research objectives (the production of new knowledge and future scientists).

Five management areas are critical to academic research projects. Procurement, communications and time management require project planning whereas people (complexity management) and project (uncertainty management) characteristics demand that a project *should not be entirely ex ante planned*. These two latter aspects are not documented in the PMBOK (PMI, 2008).

Procurement dictates planning, particularly for field investigations (data, participants, sampling) and laboratory experiments (animals, cell cultures, etc.). Planning is helpful

given that the procurement of these resources is expensive, sometime seasonal and always time consuming. Procurement even determines the extent of the period that should be *ex ante* planned in detail. *External communications* requirements also require project planning since neither communications with funding organisations, nor the publication of research results can be haphazard. This aspect is closely related to *time management* because projects must produce results at the right moment in the interest of the PI's career, student graduations and grants renewals.

Conversely, *academic organisational complexity* discourages detailed *ex ante* project planning for the following reasons:

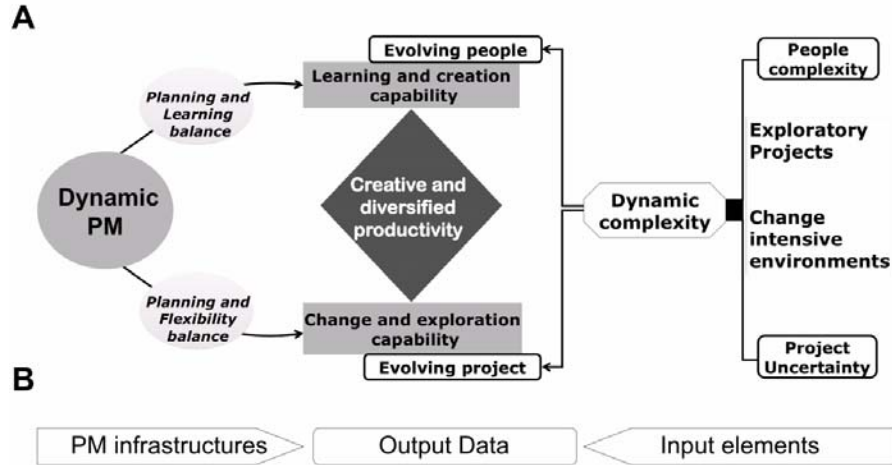
- 1 Students are partners and cannot be made to act as project executors since it would undermine their interest and compromise productivity.
- 2 Projects must be adapted to the expertise of students to benefit from their diversity.
- 3 Student diversity (background, scientific maturity, pace of learning and capacity for coping with uncertainties) render detailed *ex ante* planning counter-productive; such planning would also put them under chronic stress.
- 4 Even if PIs could entirely plan their projects, *the effort would not be beneficial* since they want students to contribute new ideas and explore promising avenues. For this to occur, it becomes necessary *not to plan* in order to allow room for some uncertainty, source of learning experience, creativity and discovery.

The uncertainties (conceptual and/or technical) characterising most research projects also discourage *ex ante* entirely planned projects. Even with well-developed methodologies as in the social sciences, field investigations require adjustments *down the line*. Chronic financial uncertainties also discourage long-term project planning. As noted, one cannot accurately predict how students and projects will evolve together, which creates organisational uncertainties. Furthermore, some research hazards are not entirely negative, since they can lead to unexpected discoveries (phenomenon of serendipity). Finally, the dynamism of the academic environment as well as PI multitasking (teaching, research and administration) are sources of many unexpected events that require continual 'micro-adjustments'.

The two aspects of PI managerial style – structure and flexibility, and their connection to organisational and technical contingencies and planning dimensions – constitute the building blocks of our model.

4.2.3 Research PM modelling

The grounded analysis suggests that in highly dynamic, uncertain and complex environments centred on discovery and operating in a context of *chronic budgetary restriction* and *time constraints*, a *dynamic type of PM*, both structured and flexible, is necessary. This dynamic PM arbitrates between *project planning and flexibility needs* and between *social planning and learning values* to support the emergence of project-people *interaction* that guarantees a *continuous flow of creative and diversified productivity*. This model is presented in Figure 4.

Figure 4 (a) A grounded model that emerges from the study and (b) its three parts

4.2.3.1 Input elements

Input elements correspond to the primary resources of the project seen as a system: the people and the project itself which both represent the system-critical elements respectively *soft* (organisational) and *hard* (technical). The interaction between people and project creates a state of *dynamic complexity*. This state is complex because the interactions are multiple and non-linear (must satisfy various interests, needs and agendas) and dynamic, since the two system components co-evolve towards *ex ante* unpredictable new states. Dynamic complexity is unavoidable (and desirable) insofar it is a prerequisite for achieving people- and project-related objectives.

4.2.3.2 PM supportive infrastructures

However, for dynamic complexity to produce results, it requires the support of dynamic PM infrastructures designed to integrate *soft* and *hard* characteristics by combining structured and flexible approaches. Dynamic PM thus provides the infrastructure for *learning and creativity* by striking a balance between training and project planning, i.e., by preserving room for uncertainty. It provides also the infrastructure needed for *exploration and for making changes to reduce project uncertainties*, by balancing project planning and managerial flexibility, i.e., by preserving the possibility of changing plans. This dynamic balance emphasises *mixed PM*, where project planning plays a strategic role in both structure (e.g., objectives, milestones, project monitoring and evaluation) and flexibility (allowing room for emerging mid-course mini-cycle planning conducive to exploration and learning).

4.2.3.3 Output data

The PM model described above provides a framework allowing the co-evolution of people and projects, and the achievement of initial project objectives. This calls attention

to the two deliverables for a PM strategy applied to exploratory, complex and uncertain projects: creative and diversified productivity (*exogenous deliverable*) and knowledge generated by learning (*endogenous deliverable*).

5 Discussion and implications

5.1 Compatibility and specificity

The results highlight clear similarities between the general management framework of research activities and classical project life cycle management whether in biology or the social sciences, for both experimental and field approaches. Furthermore, our work confirms that PIs recognise the value of PM practices, and that structured and standardised scientific methodologies are *a priori* ‘management compatible’ (Ernø-Kjølhed et al., 2001). However, these results also show that academic research projects should not be managed based *solely* on the processes, tools and techniques of classical PM because the traditional corpus of knowledge has difficulty in accommodating the uncertainty inherent in research projects, the complexity of their human resources and their dynamic interactions. The fact that research project teams include members with different expertise, potential and agendas generates *ontological uncertainty* (Bonifati, 2010), where the interaction between people and projects becomes a potential, although unpredictable, source of discovery and innovation. In addition, learning is an explicit output of academic research, and there can be no learning or capacity building (even for accomplished scientists) without room for testing options or trial-and-error approaches, which are in themselves strategies for managing research activities, especially for experimentation in biology, biochemistry or similar disciplines (Sommer and Lock, 2004). Academic research acts as an incubator for future scientists. Therefore, not only the inherent uncertainty of research projects but also the training and learning of project team members preclude ‘hyper’ planning and control.

Research projects are social systems involving system dynamics (Cooke-Davies et al., 2007) that shape the project’s future and future of project team members.

5.2 Theoretical implications

It comes as no surprise that despite the compatibility of PM principles, a contingent approach is key to implementing its processes, tools and techniques. The re-thinking of PM practices to accommodate contingency has been a trend in recent management literature (Cicmil et al., 2006). This is because PM has progressively been invading non-traditional industries for the last two or three decades. Agile PM is an example in the software sector (Fernandez and Fernandez, 2008, 2009), but current findings show what Saynisch (2010) describes as PM-2, i.e., PM in which processes combine flexibility and structure (see also Batra et al., 2010; Boehm and Turner, 2003; Styhre and Börjesson, 2011).

Academic research could be considered an ideal laboratory for exploring the management of complex and uncertain projects that focus on producing non-defined outputs in a dynamic context. Clearly, such projects should be managed differently. Classical PM, in fact, does not recognise the need to integrate dynamic complexity, flexible learning structures or flexible exploratory structures and, consequently, the

search for a balance between structure and flexibility. However, ‘atypical’ approaches that emphasise agility and flexibility only will not be helpful to these types of project. The *integrated, dynamic and evolving* combination of the two approaches during project implementation and throughout the project life cycle (see this section’s last paragraph) is what would yield optimal project performance. This is the main theoretical contribution of our work.

Perhaps another key finding is the vision of projects as systems, thus emphasising that people and technical inputs should be considered equally important (Cicmil et al., 2006; Kapsali, 2011). Managing such systems implies fostering processes that enhance the productive interaction of both. While this may seem only logical if not self-evident, the dissociation between strategic levels and operations, and divergent stakeholder agendas, may explain why problems emerge when straight and strict PM processes, disconnected from people and daily activities, are imposed. This project-system representation also emphasises that explicit outputs and explicit learning are both deliverables of complex and uncertain projects, a conclusion in accordance with the findings of Lenfle (2008) that exploratory projects always include these two deliverables.

Since projects and project teams are intertwined through learning, our results strongly support a role for PM in developing team-adaptive capabilities to allow individual learning and translate it into “coordinate actions, innovative solutions to problems and new routines adoption”; this is especially important when integrating team member diversity with broad autonomy (Burke et al., 2006). Building the capacity for developing team effectiveness and self-management and for delivering the proper balance between learning, exploration and production would therefore be beneficial and a possible avenue for further research.

We have shown that the capacity for adopting PM practices is *context-sensitive*, which is consistent with contingent PM. Furthermore, it seems reasonable to assume that it is also *project cycle-sensitive*, as also underscored by Vom Brocke and Lippe (2011). Flexible and dynamic PM appears relevant in the *implementation* phase of an academic research project when students enter the game. But for other industries, i.e., architectural projects, it may be beneficial to implement such practices at the *design* phase (complex and uncertain) where the output is not ‘fixed’ until the end of this phase draws near (Lizarralde et al., 2011). Therefore, our work could act as a starting point for further investigations of dynamic PM values in other sectors and or in specific phases of other categories of projects when learning, exploration and originality as well as structure are needed (see also Styhre and Börjesson, 2011). The lessons one could draw should add to the existing theoretical PM corpus.

5.3 Managerial implications

We were able to consider explicit *PM-like* practices as clearly distinct from scientific processes in managing academic research projects. The evaluation of scientist efficiency does not account for management variables or *a fortiori* PM variables (Adler et al., 2009). As researchers recognise (in their own words), they perform *PM-like* activities. When evaluating project progress, it helps to make allowances for what belongs respectively to the scientific or the management side, i.e., what falls within the talents of a scientist or a ‘manager’.

Our work validates what experienced scientists ‘instinctively’ do in terms of managerial practices in response to their complex and uncertain environment. In the

meantime, it offers them an opportunity to question their 'instinctive' practices, to improve processes when feasible, and to transfer them in their own words to students. Indeed, they foresee intangible benefits from PM implementation (less stress, more comfort for junior researchers and greater student autonomy) and tangible benefits (more time for scientific endeavours and increased productivity among young PIs and the project team).

While PM and the management of academic research projects may be compatible and valuable, achieving the proper mix of structure and flexibility is not easy at every level.

Institutional level: the cultural distance between PM and the scientific community should be shortened. This implies exposing young researchers to a dynamic PM framework built on 'research events', not on classical PM. Master's and PhD programmes could include such exposure.

Organisational level: balancing student training and productivity in academic research is a challenge. As outlined earlier, team adaptive capability is necessary for the development of team and individual self-management skills. As this is a major concern of PIs, they may want to implement and adapt collective approaches to suit their context, such as a collective vision of the overall project, collective goals, team self-evaluation mechanisms and new 'shareware' information tools (Bland and Ruffin, 1992; Burke et al., 2006; PMI, 2008).

Operational level: Addressing uncertainty can be accomplished through several flexible planning methods (Dalcher, 2000; Louafa and Perret, 2008). Risk management, which is more a question of experience, will benefit from developing the research teams' anticipation abilities (Goffin et al., 2010; Louafa and Perret, 2008).

5.4 Limits and conclusion

Our work is an exploratory attempt to describe, understand and categorise PI managerial activities and analyse them within a PM framework. However, its external validity is questionable because the PIs were not chosen randomly and their sample number is low. Our work is grounded in an analysis of the PIs' verbatim. Team members (students and/or professionals) and stakeholders in institutional or funding organisations were not interviewed. However, the lessons learned provide a foundation for further research, which would integrate improvements in methodology (triangulation, other disciplines), content (other corpus than the PMBOK, system dynamic and organisational theories) and an expanded scope of research.

Nevertheless, we were able to work our way toward a dynamic PM model by observing the managerial behaviour of practitioners not previously exposed to any 'dogmatic' PM. This supports a trend in the literature that is expected to rejuvenate PM by investigating real-life practices first, and then comparing the findings with the corpus of knowledge and relating it back to the professions (Cicmil et al., 2006). Contributions, both theoretical and applied, would result from respectful interaction between PM academics and practitioners.

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